Lichnerowicz Typer Vanishing Theorem for Mathai-Zhang Index

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A thesis submitted in partial fulfillment of the requirements for the degree of

Master of Science

June 2014

Abstract

This thesis has two parts. The first part, consisting of Chapter 1, reviews the basic background on spin geometry that is needed later, including the Dirac operator on a spin manifold and the classical Lichnerowicz vanishing theorem. The second part, consisting of Chapters 2–4, is devoted to a Lichnerowicz-type vanishing theorem on spin manifolds equipped with proper cocompact group actions. We first recall the Mathai–Zhang index [1] in this setting, then derive an explicit formula for the projection operator appearing in its definition, and finally prove a vanishing theorem for this index under a pointwise positivity assumption arising from the Lichnerowicz formula.

Keywords. Dirac operator; proper cocompact actions; Lichnerowicz-type vanishing theorem; Mathai–Zhang index.

Contents

Chapter 1

Introduction

Let M be a closed even-dimensional spin manifold with Riemannian metric g^{TM} on the tangent bundle TM. Let ∇^{TM} be the Levi–Civita connection and let

$$R^{TM} = (\nabla^{TM})^2$$

be the curvature of ∇^{TM}.

Let

$$S(TM) = S^+(TM) \oplus S^-(TM)$$

be the complex spinor bundle associated to (TM, g^{TM}), and let

$$\nabla^{S(TM)} = \nabla^{S^+(TM)} \oplus \nabla^{S^-(TM)}$$

be the induced Hermitian connection on $S(TM)$. For any vector $X \in TM$, we use $c(X)$ to denote Clifford multiplication by X on $S(TM)$.

The Dirac operator on M (cf. [2]) is defined by

$$D = \sum_{i=1}^{\dim M} c(e_i)\nabla^{S(TM)}_{e_i} : \Gamma(S(TM)) \longrightarrow \Gamma(S(TM)), \tag{1.1}$$

where $\{e_1, \ldots, e_{\dim M}\}$ is a local oriented orthonormal frame of TM. Its chiral part is

$$D^+ = D\Big|_{\Gamma(S^+(TM))} : \Gamma(S^+(TM)) \longrightarrow \Gamma(S^-(TM)).$$

Since M is closed, the operator D^+ is elliptic and Fredholm, and its formal adjoint is $D^- = (D^+)^*$. Therefore,

$$\operatorname{ind}(D^+) = \dim(\operatorname{Ker} D^+) - \dim(\operatorname{Ker} D^-). \tag{1.2}$$

Let $\widehat{A}(TM, \nabla^{TM})$ be the Hirzebruch characteristic form defined by

$$\widehat{A}(TM, \nabla^{TM}) = \det{}^{1/2}\left(\frac{\frac{\sqrt{-1}}{4\pi}R^{TM}}{\sinh\left(\frac{\sqrt{-1}}{4\pi}R^{TM}\right)}\right).$$

The $\widehat{A}$-genus of M is

$$\widehat{A}(M) = \int_M \widehat{A}(TM, \nabla^{TM}).$$

In 1963, A. Lichnerowicz established the classical vanishing theorem in [5]. We recall it in the following form.

Theorem 1.1 (Lichnerowicz)**.** *Let M be a closed spin manifold. If M admits a Riemannian metric of positive scalar curvature, then every harmonic spinor vanishes. In particular,*

$$\operatorname{ind}(D^+) = 0.$$

If moreover $\dim M \equiv 0 \pmod 4$*, then*

$$\widehat{A}(M) = 0.$$

Also in 1963, Atiyah and Singer established their index theorem; see [6].

Theorem 1.2 (Atiyah–Singer)**.** *Let M be a closed even-dimensional spin manifold, and let D^+ be the chiral Dirac operator on M. Then*

$$\operatorname{ind}(D^+) = \widehat{A}(M).$$

Combining Theorems 1.1 and 1.2, one sees that positive scalar curvature forces the index of the Dirac operator to vanish.

The purpose of this thesis is to establish an extension of this classical vanishing phenomenon to the case of noncompact manifolds carrying proper cocompact group actions. More precisely, we work in the unimodular setting and prove a vanishing theorem for the Mathai–Zhang index introduced in [1]. The main theorem is stated as Theorem 4.1.

The exposition is organized as follows. Chapter 2 recalls the definition and basic Fredholm property of the Mathai–Zhang index. Chapter 3 gives an explicit analytic formula for the orthogonal projection operator appearing in that construction. Chapter 4 uses the Lichnerowicz formula together with a maximum-principle argument to prove the desired vanishing theorem.

Remark 1.3. From a later bibliographical point of view, the unimodular vanishing theorem proved in Chapter 4 was subsequently generalized beyond the unimodular case; see [8, 9]. We do not pursue those developments here, since the present thesis is devoted to the unimodular setting.

Chapter 2

The Mathai–Zhang index on spin manifolds

Throughout this chapter, the action of G on M is a left action, and all integrals over G are taken with respect to a fixed left Haar measure dg.

Let M be a noncompact even-dimensional spin manifold, and let G be a locally compact group acting on M. We assume that the action is *proper* and *cocompact*, where by proper we mean that the map

$$G \times M \longrightarrow M \times M, \qquad (g, x) \longmapsto (x, gx),$$

is proper, while by cocompact we mean that the quotient space M/G is compact. We also assume that the G-action preserves the spin structure on M.

Let g^{TM} be a Riemannian metric on TM. By averaging over the group action, one may assume that g^{TM} is G-invariant (cf. [1, (2.3)]). Let

$$S(TM) = S^+(TM) \oplus S^-(TM)$$

be the complex spinor bundle associated to (TM, g^{TM}). The G-action on M lifts to an action on $S(TM)$ preserving the canonically induced Hermitian metric $g^{S(TM)}$ and Hermitian connection $\nabla^{S(TM)}$.

Let E be a Hermitian vector bundle over M equipped with a lifted G-action, a G-invariant Hermitian metric g^E, and a G-invariant Hermitian connection ∇^E. The tensor product $S(TM) \otimes E$ then carries the induced G-equivariant Hermitian metric and connection, still denoted by $g^{S(TM)\otimes E}$ and $\nabla^{S(TM)\otimes E}$.

The twisted Dirac operator is defined by

$$D^E = \sum_{i=1}^{\dim M} c(e_i)\nabla_{e_i}^{S(TM)\otimes E} : \Gamma(S(TM) \otimes E) \longrightarrow \Gamma(S(TM) \otimes E), \tag{2.1}$$

where $\{e_1, \ldots, e_{\dim M}\}$ is a local oriented orthonormal frame of TM. Its chiral parts are

$$D_{\pm}^{E} : \Gamma(S^{\pm}(TM) \otimes E) \longrightarrow \Gamma(S^{\mp}(TM) \otimes E).$$

For compactly supported smooth sections $s_1, s_2 \in \Gamma_c^{\infty}(M, S(TM) \otimes E)$, define the L^2 inner product by

$$(s_1, s_2) = \int_M \langle s_1, s_2 \rangle_{S(TM)\otimes E} \, \mathrm{d}v_{g^{TM}}. \tag{2.2}$$

Let $\| \cdot \|_0$ be the corresponding L^2 norm, and let $\| \cdot \|_1$ be a fixed G-invariant Sobolev 1-norm. Define

$$\mathbf{H}^0(M, S(TM) \otimes E) := \overline{\Gamma_c^{\infty}(M, S(TM) \otimes E)}^{\| \cdot \|_0},$$

and let $\mathbf{H}^1(M, S(TM) \otimes E)$ be the Sobolev-1 completion of the same space.

Denote the space of smooth G-invariant sections of $S(TM) \otimes E$ by

$$\Gamma(S(TM) \otimes E)^G.$$

Since M/G is compact, there exists a compact subset $Y \subset M$ such that $G(Y) = M$ (cf. [7, Lemma 2.3]). Choose open subsets $U, U' \subset M$ such that

$$Y \subset U, \qquad \overline{U} \subset U', \qquad \overline{U'} \text{ is compact.}$$

Following [1], choose a nonnegative cut-off function $f \in C_c^{\infty}(M)$ satisfying

$$f|_U = 1, \qquad \mathrm{Supp}(f) \subset U'.$$

Define $\mathbf{H}_f^0(M, S(TM) \otimes E)^G$ and $\mathbf{H}_f^1(M, S(TM) \otimes E)^G$ to be the completions of

$$\{ fs : s \in \Gamma(S(TM) \otimes E)^G \}$$

with respect to $\| \cdot \|_0$ and $\| \cdot \|_1$, respectively. Since f has compact support, every section of the form fs is compactly supported, so these norms are well defined. By Hilbert space theory, there exists a unique orthogonal projection

$$P_f : \mathbf{H}^0(M, S(TM) \otimes E) \longrightarrow \mathbf{H}_f^0(M, S(TM) \otimes E)^G.$$

Moreover,

$$P_f D^E : \mathbf{H}_f^1(M, S(TM) \otimes E)^G \longrightarrow \mathbf{H}_f^0(M, S(TM) \otimes E)^G$$

is a well-defined operator.

We now recall the basic Fredholm property proved in [1, Proposition 2.1].

Proposition 2.1 (Mathai–Zhang)**.** *The operator*

$$P_f D^E : \mathbf{H}^1_f(M, S(TM) \otimes E)^G \longrightarrow \mathbf{H}^0_f(M, S(TM) \otimes E)^G$$

is Fredholm.

It was shown in [1] that the index of the chiral part

$$P_f D^E_+ : \mathbf{H}^1_f(M, S^+(TM) \otimes E)^G \longrightarrow \mathbf{H}^0_f(M, S^-(TM) \otimes E)^G$$

is independent of the choice of the cut-off function f, as well as of the G-invariant metrics and connections involved. Following [1, Definition 2.4], we denote this index by

$$\mathrm{ind}_G(D^E_+).$$

This is the Mathai–Zhang index of the twisted Dirac operator.

Chapter 3

Explicit formula for the projection operator P_f

In this chapter, we give an explicit formula for the projection operator appearing in Chapter 2. Since the vanishing theorem proved later is stated for unimodular groups, we assume throughout this chapter that G is unimodular.

Let $S \to M$ be a G-equivariant Hermitian vector bundle. For $g \in G$ and $x \in M$, we denote by

$$g : S_{g^{-1}x} \longrightarrow S_x$$

the induced unitary map on fibres. Equivalently, the action of G on sections is given by

$$(g \cdot s)(x) := g\Big(s(g^{-1}x)\Big).$$

A smooth section s is G-invariant if and only if $g \cdot s = s$ for all $g \in G$.

For $x \in M$, define

$$A(x)^2 := \int_G f(g^{-1}x)^2 \, dg. \tag{3.1}$$

To justify this definition, fix $x \in M$ and consider the set

$$K_x := \{g \in G : f(g^{-1}x) \neq 0\}.$$

Its inverse is

$$K_x^{-1} = \{h \in G : f(hx) \neq 0\} = \{h \in G : hx \in \mathrm{Supp}(f)\}.$$

Because the action is proper, the map

$$G \longrightarrow M \times M, \qquad h \longmapsto (x, hx)$$

is proper. Hence the inverse image of the compact set $\{x\} \times \mathrm{Supp}(f)$ is compact, so K_x^{-1} is compact, and therefore K_x is compact as well. This proves that the integral in (3.1) is finite. Since $G(U) = M$ and $f|_U = 1$, for each $x \in M$ there exists $g_0 \in G$ such that $g_0^{-1}x \in U$. As U is open and the action is continuous, one has $g^{-1}x \in U$ for all g sufficiently close to g_0, and hence $f(g^{-1}x) = 1$ on a neighborhood of g_0. It follows that the integral in (3.1) is strictly positive. Moreover, by left invariance of Haar measure,

$$A(hx) = A(x), \qquad h \in G. \tag{3.2}$$

Lemma 3.1. *The function $A : M \to (0, \infty)$ is smooth and G-invariant.*

Proof. The integrand in (3.1) is smooth in x, and for each fixed x the set of $g \in G$ on which it is nonzero is contained in the compact set K_x. By properness of the action, if x' varies in a sufficiently small neighborhood of x, then all sets $K_{x'}$ are contained in a single compact subset of G. One may therefore differentiate under the integral sign, which shows that A^2 is smooth, and hence so is A because A is strictly positive. The G-invariance follows from a direct change of variables:

$$A(hx)^2 = \int_G f(g^{-1}hx)^2\, dg = \int_G f(z^{-1}x)^2\, dz = A(x)^2.$$

□

For $s \in \Gamma_c^\infty(M, S)$, define

$$(Q_f s)(x) := \frac{f(x)}{A(x)^2} \int_G f(g^{-1}x)\, g\Big(s(g^{-1}x)\Big)\, dg. \tag{3.3}$$

The same compact-support argument as above shows that the integral is well defined for all $x \in M$ and produces a smooth section.

Proposition 3.2. *For every $s \in \Gamma_c^\infty(M, S)$, the section $Q_f s$ belongs to*

$$\{f\sigma : \sigma \in \Gamma(S)^G\}.$$

Moreover, for every $\sigma \in \Gamma(S)^G$,

$$(s - Q_f s, f\sigma) = 0. \tag{3.4}$$

Consequently, $Q_f s = P_f s$ for all $s \in \Gamma_c^\infty(M, S)$.

Proof. Define a smooth section σ_s by

$$\sigma_s(x) := \frac{1}{A(x)^2} \int_G f(g^{-1}x)\, g\Big(s(g^{-1}x)\Big)\, dg. \tag{3.5}$$

Then $Q_f s = f\sigma_s$ by definition. We first show that σ_s is G-invariant. Let $h \in G$.

Using (3.2) and the change of variables $g = hz$, one gets

$$\begin{aligned}\sigma_s(hx) &= \frac{1}{A(hx)^2}\int_G f(g^{-1}hx)\,g\Big(s(g^{-1}hx)\Big)\,dg\\ &= \frac{1}{A(x)^2}\int_G f(z^{-1}x)\,hz\Big(s(z^{-1}x)\Big)\,dz\\ &= h\left(\frac{1}{A(x)^2}\int_G f(z^{-1}x)\,z\Big(s(z^{-1}x)\Big)\,dz\right)\\ &= h\sigma_s(x).\end{aligned}$$

Hence $\sigma_s \in \Gamma(S)^G$, and therefore

$$Q_f s = f\sigma_s \in \{f\sigma : \sigma \in \Gamma(S)^G\}.$$

Next let $\sigma \in \Gamma(S)^G$. Using (3.3), the G-invariance of the Hermitian metric, Fubini's theorem, and the change of variables $x = gy$, we obtain

$$\begin{aligned}(Q_f s, f\sigma) &= \int_M \left\langle \frac{f(x)}{A(x)^2}\int_G f(g^{-1}x)\,g\Big(s(g^{-1}x)\Big)\,dg, f(x)\sigma(x)\right\rangle \mathrm{d}v_{g^{TM}}(x)\\ &= \int_G\int_M \frac{f(gy)^2 f(y)}{A(gy)^2}\langle g(s(y)), \sigma(gy)\rangle\,\mathrm{d}v_{g^{TM}}(y)\,dg\\ &= \int_G\int_M \frac{f(gy)^2 f(y)}{A(y)^2}\langle s(y), \sigma(y)\rangle\,\mathrm{d}v_{g^{TM}}(y)\,dg.\end{aligned}$$

Since G is unimodular, inversion preserves Haar measure, and hence

$$\int_G f(gy)^2\,dg = \int_G f(g^{-1}y)^2\,dg = A(y)^2.$$

Therefore,

$$(Q_f s, f\sigma) = \int_M f(y)\langle s(y), \sigma(y)\rangle\,\mathrm{d}v_{g^{TM}}(y) = (s, f\sigma).$$

This proves (3.4).

Finally, the subspace

$$\{f\sigma : \sigma \in \Gamma(S)^G\}$$

is dense in $\mathbf{H}^0_f(M,S)^G$ by definition. Since $Q_f s$ lies in this dense subspace and $s - Q_f s$ is orthogonal to it, $Q_f s$ is exactly the orthogonal projection of s onto $\mathbf{H}^0_f(M,S)^G$. Hence $Q_f s = P_f s$ for every $s \in \Gamma^\infty_c(M,S)$. □

Corollary 3.3. *The formula (3.3) agrees with the orthogonal projection P_f on the dense subspace $\Gamma^\infty_c(M,S) \subset \mathbf{H}^0(M,S)$. Consequently, it extends by continuity to all of $\mathbf{H}^0(M,S)$ and yields the orthogonal projection there as well.*

Proof. By Proposition 3.2, the operator defined by (3.3) agrees with P_f on the dense subspace $\Gamma^\infty_c(M,S)$. Since P_f is bounded, the formula extends uniquely by continuity to all of $\mathbf{H}^0(M, S)$. □

Remark 3.4. For non-unimodular groups, an analogous explicit formula involves the modular character. Since the vanishing theorem in this thesis is proved in the unimodular case, the above formula is sufficient for our purposes.

Chapter 4

A Lichnerowicz-type vanishing theorem for the Mathai–Zhang index

In this chapter, we adapt the classical Bochner–Lichnerowicz argument to the Mathai–Zhang index in the unimodular proper cocompact setting.

Let k^{TM} be the scalar curvature of the G-invariant metric g^{TM}, and let R^E be the curvature of ∇^E. If $\{e_i\}$ is a local orthonormal frame of TM, define the Hermitian endomorphism $c(R^E)$ of $S(TM) \otimes E$ by

$$c(R^E) = \frac{1}{2} \sum_{i,j=1}^{\dim M} c(e_i)c(e_j)R^E(e_i, e_j). \tag{4.1}$$

Recall the classical Lichnerowicz formula (cf. [2, Chapter II, Theorem 8.17]) for the twisted Dirac operator:

$$(D^E)^2 = -\Delta^E + \frac{k^{TM}}{4} + c(R^E), \tag{4.2}$$

where the Bochner Laplacian is given by

$$\Delta^E = \sum_{i=1}^{\dim M} \left(\left(\nabla_{e_i}^{S(TM)\otimes E} \right)^2 - \nabla_{\nabla_{e_i}^{TM} e_i}^{S(TM)\otimes E} \right).$$

We can now state the main result.

Theorem 4.1. *Let M be a noncompact even-dimensional spin manifold carrying a proper cocompact action by a locally compact unimodular group G, and assume that the action preserves the spin structure on M. Let E be a G-equivariant Hermitian vector bundle over M with a G-invariant Hermitian metric and a G-invariant Hermitian* *connection ∇^E. Assume that the Hermitian endomorphism*

$$R := \frac{k^{TM}}{4} + c(R^E) \tag{4.3}$$

is pointwise positive definite on $S(TM)\otimes E$, that is,

$$\langle R(x)u, u\rangle > 0 \qquad \text{for all } x \in M \text{ and all } u \in (S(TM)\otimes E)_x \setminus \{0\}.$$

Then

$$\operatorname{ind}_G(D^E_+) = 0.$$

Proof. Since G is unimodular, Mathai–Zhang proved in [1, Theorem 2.7] that

$$\operatorname{ind}_G(D^E_+) = \dim\Big((\operatorname{Ker} D^E_+)^G\Big) - \dim\Big((\operatorname{Ker} D^E_-)^G\Big). \tag{4.4}$$

Therefore it is enough to prove that

$$(\operatorname{Ker} D^E_+)^G = (\operatorname{Ker} D^E_-)^G = \{0\}.$$

Let $s \in (\operatorname{Ker} D^E_\pm)^G$. By elliptic regularity, s is smooth. Since $D^E s = 0$, the Lichnerowicz formula (4.2) gives

$$(D^E)^2 s = 0.$$

Fix any $\varphi \in C^\infty_c(M)$. Pairing $(D^E)^2 s$ with φs and integrating, we get

$$\begin{aligned}
0 &= \int_M \langle (D^E)^2 s, \varphi s\rangle \, \mathrm{d}v_{g^{TM}} \\
&= \int_M \Big(\langle -\Delta^E s, \varphi s\rangle + \varphi \langle Rs, s\rangle \Big) \, \mathrm{d}v_{g^{TM}} \\
&= \int_M \left(\sum_{i=1}^{\dim M} \langle \nabla_{e_i} s, \nabla_{e_i}(\varphi s)\rangle + \varphi\langle Rs, s\rangle \right) \mathrm{d}v_{g^{TM}} \\
&= \int_M \left(\frac{1}{2} \sum_{i=1}^{\dim M} e_i(\varphi) e_i(|s|^2) + \varphi |\nabla s|^2 + \varphi\langle Rs, s\rangle \right) \mathrm{d}v_{g^{TM}}.
\end{aligned} \tag{4.5}$$

Since the left-hand side vanishes for every compactly supported test function φ, it follows that the smooth function

$$\frac{1}{2}\Delta(|s|^2) - |\nabla s|^2 - \langle Rs, s\rangle$$

vanishes identically on M. Hence

$$\frac{1}{2}\Delta(|s|^2) = |\nabla s|^2 + \langle Rs, s\rangle \geq \langle Rs, s\rangle. \tag{4.6}$$

Choose a compact subset $Y \subset M$ such that $G(Y) = M$; such a subset exists because M/G is compact. Since s is G-invariant, the function $|s|^2$ is constant along G-orbits. Hence any maximum of $|s|^2$ on the compact set Y is automatically a global maximum on all of M. Let $p \in Y$ be such a point. By the maximum principle,

$$\Delta(|s|^2)(p) \leq 0.$$

Evaluating (4.6) at p, we obtain

$$0 \geq \frac{1}{2}\Delta(|s|^2)(p) = |\nabla s|^2(p) + \langle R(p)s(p), s(p)\rangle \geq \langle R(p)s(p), s(p)\rangle \geq 0.$$

Since $R(p)$ is positive definite, it follows that $s(p) = 0$. As p is a maximum point of the nonnegative function $|s|^2$, we conclude that $|s|^2 \equiv 0$ on M, hence $s \equiv 0$.

Thus both $(\operatorname{Ker} D_+^E)^G$ and $(\operatorname{Ker} D_-^E)^G$ are trivial. By (4.4), this implies

$$\operatorname{ind}_G(D_+^E) = 0.$$

The proof is complete. $\square$

Corollary 4.2. *Let M be a noncompact even-dimensional spin manifold carrying a proper cocompact action by a locally compact unimodular group G, and assume that the action preserves the spin structure on M. If M admits a G-invariant Riemannian metric of positive scalar curvature, then*

$$\operatorname{ind}_G(D_+) = 0.$$

Proof. This is the special case of Theorem 4.1 obtained by taking E to be the trivial line bundle with its flat connection, so that $c(R^E) = 0$. $\square$

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